

Deep Learning Based Ovarian Ultrasound Image Classification Using Supervised and Semi-Supervised Learning

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Accurate diagnosis is crucial for managing common ovarian illnesses such as polycystic ovary syndrome (PCOS) and complex cysts. Ultrasound imaging is a common diagnostic tool, however interpretation can be problematic due to picture noise and inadequate labeled data. This paper proposes a deep learning algorithm to automatically classify ovarian ultrasound images as healthy, PCOS, or complex cysts. Additional key words and phrases include ultrasound imaging, ovarian cyst classification, PCOS, deep learning, ResNet, supervised learning, and semi-supervised learning.

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1 Research Questions

- Can deep learning models accurately classify ovarian ultrasound images into healthy, PCOS and complicated cyst categories?
- How does semi-supervised learning (with 10% labeled and 90% unlabeled data) compare to fully supervised learning (with only 90% labeled data) in classification performance?

2 Introduction

2.1 Clinical Significance and Background

Ovarian problems, such as Polycystic Ovary Syndrome (PCOS) and various forms of ovarian cysts, are among the leading causes of reproductive health issues in women worldwide. Accurate identification of these disorders is critical for effective clinical care and long term health consequences. Ultrasound imaging is the diagnostic standard due to its non-invasive nature, low cost, and real time imaging capabilities.

2.2 Challenges of Labeled Data in Generative AI in Medicine

Deep learning techniques, notably Convolutional Neural Networks (CNNs), have shown enormous potential for automating medical image classification. However, the performance of these models is usually dependent on the availability of huge, carefully annotated datasets. In the medical domain, getting such labels is a substantial bottleneck; manual annotation takes a long time from qualified physicians and is frequently prohibited by privacy rules. This "data scarcity" issue commonly leads to overfitting, in which a model memorizes a tiny training set but fails to generalize to the different imaging devices and patient groups encountered in real world clinical settings.

3 Related Work

The training methodology used in this study is based on recognized approaches for medical image analysis, which emphasize the importance of deep feature extraction over traditional manual engineering. Following Suha and Islam's findings (2022), we used a transfer learning strategy with an NVIDIA-optimized ResNet50 on large scale datasets.

This decision is confirmed by He et al. (2024), who found that Residual Networks (ResNet) regularly perform comparable to expert subjective assessments (SA) while evaluating complex ovarian structures. To optimize the learning process, we used the Adam algorithm, which is consistent with recent research by Sood et al. (2025), who used adaptive gradient descent methods to address the high level of noise and fluctuating contrast levels found in ultrasound data.

4 Methods

4.1 Datasets

4.1.1 Data Sources and Class Distribution. The dataset used in this study consists of ovarian ultrasound images collected from multiple publicly available sources. To improve diversity and effectiveness, these datasets were merged into a single unified dataset. The final dataset includes four classes: healthy ovaries, polycystic ovary syndrome (PCOS), cysts, and complicated cysts.

Combining multiple datasets introduces variability in imaging conditions such as resolution, noise, and acquisition techniques, which helps improve the model's ability to generalize across different clinical scenarios.

4.1.2 Data Preprocessing. Before training, all images were standardized through preprocessing steps to ensure consistency across the merged dataset. This included resizing images to a fixed resolution and normalizing pixel values.

In addition, data augmentation techniques such as rotation and flipping were applied to increase variability in the dataset and reduce overfitting. These preprocessing steps are essential for improving model performance, especially when working with limited labeled medical data.

4.1.3 Dataset Splitting Strategy. The dataset was divided following the structure illustrated in Fig. 1. First, the full dataset was split into 90% training and validation data and 10% testing data, where the test set was kept completely unseen during training.

The training and validation portion was further divided into 80% training data and 10% validation data. Within the training set, only 10% of the data was labeled, while the remaining 90% was treated as unlabeled data. This setup simulates real world conditions where labeled medical data is limited.

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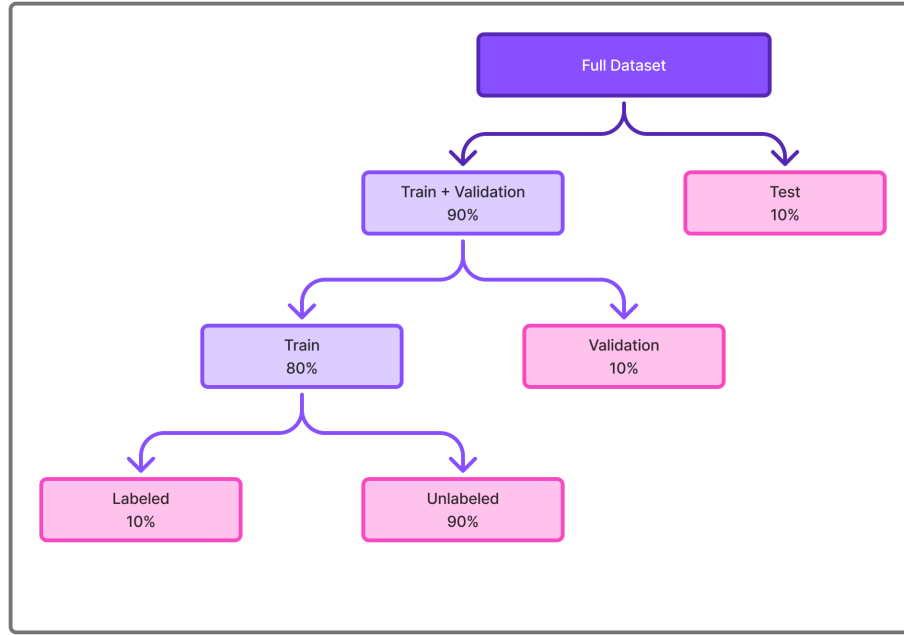


Fig. 1. Data Splitting Scheme.

4.1.4 Data Preparation and Format. To prepare the dataset for model training, the image files were first organized and compressed into a zipped format for efficient storage and uploading into the model environment. Afterward, the dataset was converted into a CSV file format, where each row contains the image path and its corresponding label for labeled data. This format ensures compatibility with the training pipeline and allows for easier data handling during both supervised and semi-supervised learning stages.

4.1.5 Semi-Supervised Learning Setup. The labeled portion of the training data (10%) was used to train an initial supervised model. Once trained, this model was applied to the unlabeled portion (90%) to generate pseudo-labels. These pseudo-labeled data points were then incorporated into the training process using a semi-supervised learning approach. This method allows the model to learn from a significantly larger dataset without requiring additional manual labeling, improving overall performance and generalization.

5 Experiment

5.1 Proposed Methodology

5.1.1 Model Architecture and Optimization. Our classification process is built around the NVIDIA optimized ResNet50 architecture (Figure 2 shows our implementation). We chose this model because of its demonstrated efficacy in medical feature extraction, particularly in identifying the complicated, nonlinear patterns of ovarian follicles and cysts. To tailor the model to our unique diagnostic objective, we substituted the final fully linked layer with a custom classification head for four distinct classes: healthy, PCOS, ovarian cyst, and complicated.

For the training process, we utilize the Adam optimizer with an initial learning rate of 0.001. This choice is grounded in Adam's ability to handle the "noise" typical of ultrasound imagery by maintaining adaptive learning rates for each parameter, ensuring stable convergence even with the varied intensities found in our merged datasets.

5.1.2 The Semi - Supervised Training Approach. This technique greatly increases the model's robustness by enabling it to learn from a large pool of unlabeled data.

We train an initial model using a small, expert annotated set (10% of the total data). This stage establishes the fundamental feature extraction capabilities of the ResNet50 architecture. One problem we had to overcome was the model memorizing the data. Because we used only 10% of the data, there were more chances of overfitting. The model then performs inference on the remaining 90% of unlabeled images. We apply a strict Confidence Threshold ($T \geq 0.90$) to the softmax output. Only images where the model is highly certain are assigned "Pseudo-Labels."

5.2 Experimental Setup

5.2.1 Computational Environment. All experiments in this study were conducted using Google Colab, which provides a cloud-based environment with access to GPU acceleration. Google Colab was selected due to its ease of use, availability of preinstalled deep learning libraries, and ability to handle large scale image datasets without requiring local computational resources.

5.2.2 Data Upload and Extraction. The dataset was prepared locally and compressed into a zipped file format for efficient uploading. The zipped dataset was then uploaded into the Google Colab environment. Within the notebook, the dataset was extracted using standard file handling commands, allowing direct access to image files during training. This approach simplified dataset management and ensured that all data was organized within the runtime environment.

5.2.3 Training Pipeline Setup. After extraction, the dataset was loaded into the training pipeline using the CSV file created during preprocessing. The CSV file provided structured access to image

paths and labels, enabling efficient data loading and batching during training.

The labeled portion of the dataset was first used to train the initial model. The trained model was then applied to the unlabeled data to generate pseudo-labels, which were incorporated into the semi-supervised learning process.

5.2.4 Model Training Configuration. The training process was implemented using deep learning frameworks available in Google Colab. Standard training techniques such as batch processing, validation monitoring, and early stopping were used to ensure stable learning. The validation set was used to monitor performance during training, while the test set was reserved for final evaluation to ensure unbiased results.

5.3 Training Performance and Model Fitting Analysis

The training and validation graphs provide important insight into the model's fitting behavior, particularly in terms of underfitting and overfitting.

In the early training stages, the model exhibits signs of underfitting, where both training and validation accuracy are relatively low and the loss values remain high. This occurs because the model is initially trained on only a small portion of labeled data (10%), limiting its ability to capture meaningful patterns in the dataset. The slow increase in accuracy and gradual decrease in loss reflect this limited learning capacity at the beginning.

As training progresses and pseudo-labeled data is incorporated, the model begins to learn more effectively. The training accuracy

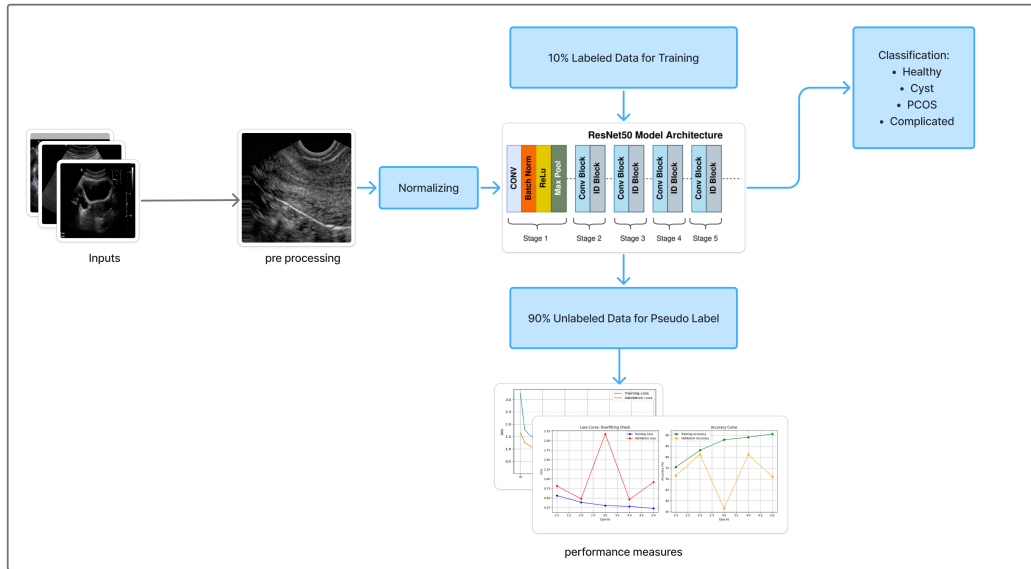


Fig. 2. Proposed Semi - Supervised Training Architecture.

increases significantly, and the training loss decreases steadily, indicating improved feature extraction and pattern recognition. This transition shows that the semi-supervised approach helps overcome early underfitting by effectively expanding the dataset.

However, the graphs also reveal mild signs of overfitting (See figure 3). Specifically, a gap begins to appear between the training and validation curves, where the training accuracy continues to improve while the validation accuracy stabilizes or fluctuates. Similarly, the training loss decreases more rapidly than the validation loss. This suggests that the model starts to fit the training data more closely than the unseen validation data.

This overfitting behavior can be attributed to the inclusion of pseudo-labeled data, which may contain incorrect or noisy labels. As a result, the model may learn patterns that are not fully generalizable. Despite this, the overfitting observed is not severe, as the validation performance remains relatively stable and does not show significant degradation.

Overall, the graphs indicate that the model successfully transitions from underfitting to a well-trained state, with only moderate overfitting. The use of techniques such as dropout, validation monitoring, and early stopping helps control overfitting and maintain generalization performance.

6 Results

The model demonstrated promising early-stage performance, with qualitative validation showing an 83.3% match rate on the initial randomized test batch (5 out of 6 images correctly classified), as shown in Table 1 and Figure 4. The single misclassification occurred on a PCOS sample, likely due to the visual similarity between PCOS and other ovarian conditions in ultrasound imagery.

As seen in Figure 5, the model-generated pseudo-labels across a broader set of unlabeled samples appear largely consistent with expected class characteristics, though a tendency toward "healthy"

predictions suggests a potential class imbalance effect in the pseudo-labeling stage. The training curves in Figure 3 reflect a pattern consistent with early underfitting that improves as pseudo-labeled data is incorporated, followed by a mild overfitting trend indicated by the growing gap between training and validation accuracy. This is expected given that the model was initially trained on only 10% labeled data, and pseudo-labels may carry some noise.

7 Challenges and Limitations

Even though the semi-supervised approach helps with the lack of labeled data, there are still some challenges we ran into.

One challenge is the nature of ultrasound images themselves. They tend to be noisy, low in contrast, and sometimes hard to interpret. On top of that, since we combined data from multiple public datasets, the images vary in quality, resolution, and how they were taken. This makes preprocessing harder and can sometimes cause the model to miss important visual details.

Another issue is with pseudo-labeling. Even with a strict confidence threshold, the model can still be wrong about some images while being very confident. When those wrong labels get added to the training data, the model may learn from those mistakes and struggle with borderline cases, like telling apart a complicated cyst from an irregular follicle.

Lastly, there are computational limitations. Since we used Google Colab, we were limited in GPU memory and runtime. Processing thousands of unlabeled images through ResNet50 is heavy on resources, which made it difficult to experiment with different settings

8 Conclusion

This midterm report presents the progress made toward developing a deep learning model for ovarian ultrasound image classification. The dataset was successfully constructed by merging multiple sources and organizing the data into four classes: healthy, PCOS, cyst, and

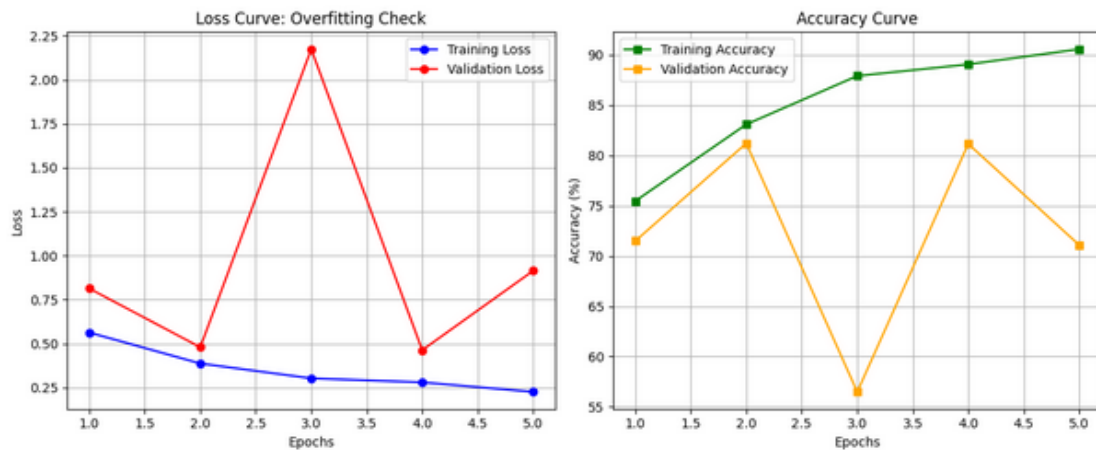


Fig. 3. Training Stats Graph.

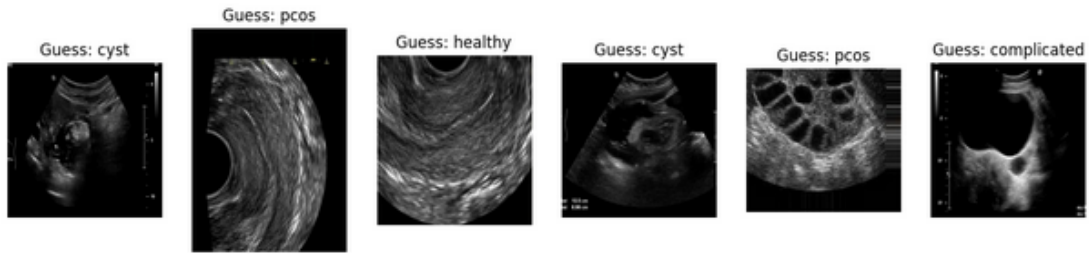


Fig. 4. Randomized sample of six ultrasound images for qualitative validation.

Table 1. Qualitative Validation of Pseudo-Labeled Ovarian Ultrasound Samples

Sample ID	Predicted Class	Match
Test 1: Initial Randomized Batch		
Image_01	Simple Cyst	Yes
Image_02	PCOS	No
Image_03	Healthy	Yes
Image_04	Simple Cyst	Yes
Image_05	PCOS	Yes
Image_06	Complicated	Yes

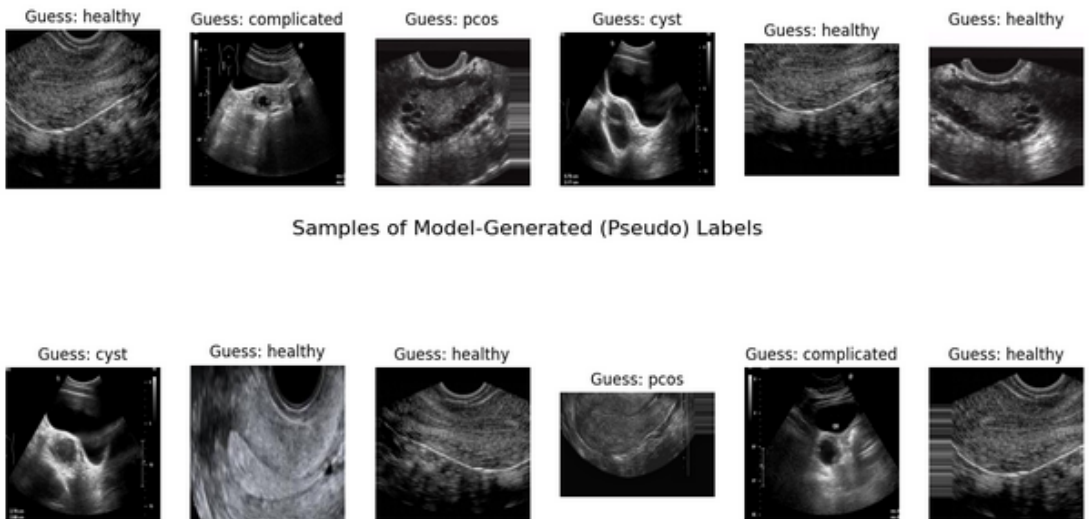


Fig. 5. More randomized pseudo - labels.

complicated cyst. A structured data splitting strategy was implemented, along with preprocessing and augmentation techniques to improve data quality and consistency.

The experimental setup was established using Google Colab, where

the dataset was uploaded, extracted, and prepared in CSV format for training. A semi-supervised learning approach was initiated by training the model on a limited labeled subset (10%) and generating pseudo-labels for the remaining unlabeled data. Initial training

results indicate that incorporating unlabeled data can improve learning efficiency and model performance compared to using labeled data alone.

At this stage, the work is focused on the training phase, and the full model evaluation and optimization are still in progress. Future work will include refining the semi-supervised learning process, improving pseudo-label accuracy, conducting comprehensive testing on the unseen dataset, and comparing different model architectures. These steps are expected to provide a more complete assessment of the model's performance and its potential for real-world medical applications.

9 Future Work

For the remainder of the project, we plan to complete the full evaluation of the model on the held-out test set, including accuracy, precision, recall, F1-score, and confusion matrices across all four classes.

We also plan to run a fully supervised baseline using 90% labeled data so we can directly compare it against our semi-supervised approach and answer our second research question. This comparison is central to the goals of the project.

One area we want to focus on is improving the quality of the data itself. Since the dataset was built by merging multiple public sources, there is a lot of variability in image quality. Better preprocessing, stricter filtering of low-quality images, and more careful class balancing could significantly improve model performance. As the saying goes, garbage in, garbage out, no matter how good the model is, it can only be as good as the data it learns from. To address the overfitting observed in the current results, we also intend to experiment with stronger regularization techniques and explore ways to improve the quality of pseudo-labels, such as adjusting the confidence threshold or filtering out uncertain predictions better.

If time and resources allow, we would also like to test the model on a completely external dataset to better evaluate how well it generalizes beyond the data it was trained on.

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